One refusal neuron, across languages

Progress since 22 October: Arabic, 22 more language sets, and the refusal direction inside the MLP

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Where we were, and the three parts we agreed on

22 October: Apple's single refusal neuron replicated in English on Qwen3-8B, Llama-3.1-8B and Qwen3-1.7B. Our search finds their exact neurons; pinning one jailbreaks the model (90–97%). First sign that the same neuron fires on Arabic.

Part 1: the Apple paper in Arabic

- Every experiment of the paper, in Arabic, on two datasets (PolyRefuse, MultiJail)
- Plus the English gaps

Mostly done

Part 2: every other language

- PolyRefuse: 14 languages
- MultiJail: 10 human-translated languages
- One neuron for all languages, per family, or per language?

Search done · attacks running

Part 3: MLP questions

- Nanda's direction built inside the MLP
- How much of it is the neuron?
- Does it work only through the neuron?
- Reverse route: direction first, then its top neuron

Done in English · Arabic partly

Setup used throughout

Model	Apple's neuron	Pin used
Qwen3-8B	layer 14, neuron 7924	+20
Llama-3.1-8B-Instruct	layer 11, neuron 4258	−4
Qwen3-1.7B	layer 13, neuron 3270	+40

JUDGES

- English: Llama-Guard-3
- Other languages: Qwen3Guard (reads the language) and Llama-Guard on an NLLB-200 translation
- Apple's main criterion (App. K.1): attack succeeds if the answer is not a refusal and is coherent; judged by gemma-2-9b-it

DATA

- English: JailbreakBench (100) and HarmBench (191)
- PolyRefuse: 14 languages, the same harmful prompts translated
- MultiJail: 10 languages, human-translated, more borderline prompts
- Per language: search on 128 harmful / 128 harmless, test on held-out prompts

INTERVENTIONS

- Constant pin: the neuron held at the pin at every token
- Anchor k=1, 2: pin scaled by how strongly the neuron fires on that prompt, so harmless prompts are barely touched (k = strength)
- Nanda baseline: refusal direction removed from the residual stream

In Arabic, the same neuron is found, and pinning it jailbreaks

- Apple's search run from scratch on Arabic prompts finds Apple's English neuron again (Qwen3-8B, Llama; near-tie on Qwen3-1.7B)
- The neuron fires on Arabic harmful prompts at 60–80% of its English strength, at the same template token

SUCCESSFUL ATTACKS, ARABIC, 100 HELD-OUT PROMPTS (APPLE'S K.1 CRITERION)

Model	PolyRefuse: no pin → pinned	MultiJail: no pin → pinned	Nanda, Arabic direction (PolyRefuse)
Qwen3-8B	**15% → 97%**	15% → 82%	97%
Llama-3.1-8B	**22% → 96%**	30% → 91%	99%
Qwen3-1.7B	**32% → 99%**	39% → 92%	100%

Arabic behaves like English. One neuron switches refusal off as well as Nanda's whole direction does, and every pinned answer stays coherent.

Arabic: small cost to ability, and the neuron is a good detector

ARABIC MMLU ACCURACY (1,000 QUESTIONS)

Model	No pin	Constant	Anchor k=1	Nanda
Qwen3-8B	56.4	46.9	**55.7**	55.3
Llama-3.1-8B	45.3	40.5	**45.3**	44.0
Qwen3-1.7B	39.5	31.7	**38.4**	36.4

DETECTING HARMFUL PROMPTS (AUROC)

Detector	Arabic	English
The neuron (3 models)	**0.95–0.995**	0.98–1.00
Llama-Guard-3	0.995	1.00

- Anchored pin (Apple's gentler version) costs about 1 point; a constant pin costs 5–9
- One number from one neuron detects harmful Arabic prompts almost as well as a dedicated 8B safety model
- Apple's neuron is the 1st or 2nd most aligned neuron with Nanda's Arabic refusal direction

Every Arabic result so far matches the English paper. Remaining: what the neuron responds to (needs a human reader).

English gaps filled: the neuron beats the residual stream

ATTACK SUCCESS OF THE BEST SINGLE DIMENSION (K.1 CRITERION)

Model	MLP neuron	Residual stream
Llama-3.1-8B	**97%**	32%
Qwen3-8B	**96%**	16%

Apple reports 29–45% (Llama) and 39% (Qwen3-8B) for the residual stream.

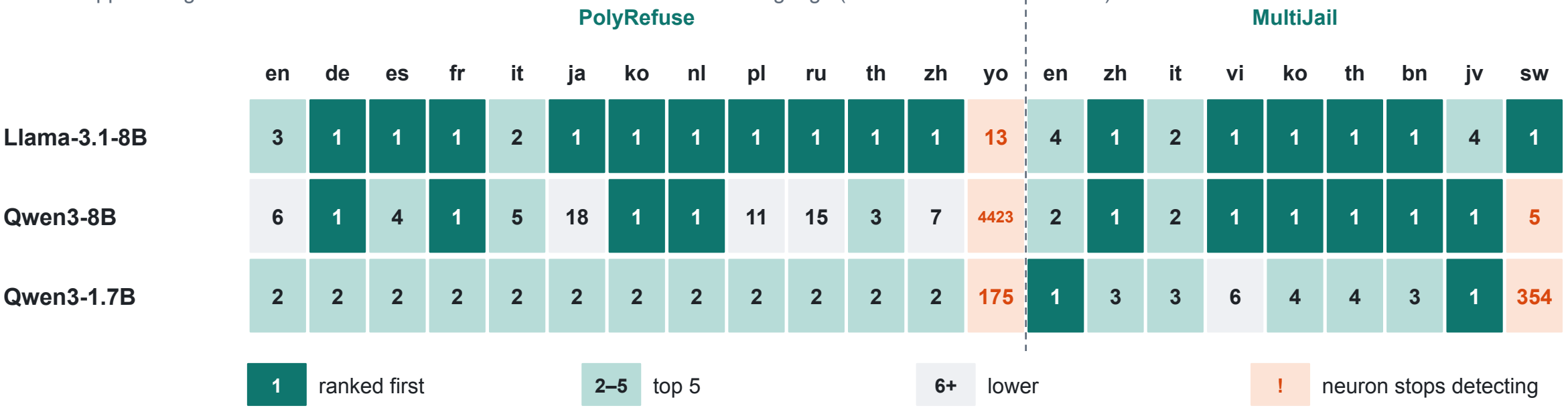
ARDITI'S ABLATION: COST TO ABILITY (ENGLISH)

Model	MMLU	GSM8K
Qwen3-8B	67.5 → 66.6	58.0 → 57.2
Llama-3.1-8B	63.8 → 63.5	72.8 → 74.4
Qwen3-1.7B	53.7 → 51.7	58.4 → 57.6

Section 3.3 confirmed: the refusal switch lives in one MLP neuron, not in any single residual-stream dimension. Nanda's ablation costs 1–2 points.

Every language: the same neuron comes out on top

Rank of Apple's English neuron when the search is rerun from scratch in each language (out of ~110k–300k neurons)



	In the top 5	Ranked first	Detection AUROC (other languages)
Llama-3.1-8B	**21 of 22**	16 of 22	0.85–1.00
Qwen3-8B	16 of 22	10 of 22	0.95–0.99
Qwen3-1.7B	19 of 22	2 of 22 (2nd in 12)	0.92–0.99

Across languages it is one small shared group of neurons

LLAMA-3.1-8B

Neuron	Top-5 lists (of 22)
L11:F4258 (Apple)	21
L15:F9635	18
L10:F1084	16
L10:F4789	15
L10:F9064	13

QWEN3-8B

Neuron	Top-5 lists (of 22)
L14:F7924 (Apple)	16
L17:F11229	16
L16:F8474	13
L16:F5318	12
L18:F7664	11

QWEN3-1.7B

Neuron	Top-5 lists (of 22)
L13:F3270 (Apple)	19
L10:F244	19
L12:F4580	17
L15:F1109	15
L12:F2897	11

- The same few neurons are found whether the search runs in English, German, Thai, Bengali or Chinese
- Llama has one dominant refusal neuron for all languages; the Qwen models have a group of 2–5 neurons
- Part 3 finds the same group from a different starting point (slide 12)

> **Answer to Part 2's main question:** one shared, language-agnostic refusal circuit, not one neuron per language.

Exceptions: in Yoruba and Swahili the refusal neuron goes quiet

	Model refuses (no pin)	Neuron detects harm (AUROC)	Apple's neuron rank
Other languages	51–98%	0.85–1.00	1–18
Yoruba	10–44%	**0.60–0.81**	13 · 175 · 4423
Swahili (Qwen models)	43–55%	**0.34–0.65**	5 · 354

WHAT IT SUGGESTS

- In these languages the harmful request does not reach the refusal circuit, so the model never refuses
- A mechanistic explanation of low-resource-language jailbreaks: the attack works because the refusal neuron stays silent

POSSIBLE DIRECTION

- Attack: rank languages by how weakly the neuron fires, to predict which translations jailbreak
- Defense: pin the neuron to its harmful value on low-resource prompts and see whether refusal comes back

The neuron doubles as a gauge of whether a language is 'understood' as harmful. Proposed as a jailbreak direction.

PART 2: EVERY OTHER LANGUAGE

Does pinning the neuron switch refusal off in every language?

Search finds the neuron (correlation). This test pins it and checks that refusal stops (cause), in each language.

QWEN3-1.7B, 100 HELD-OUT PROMPTS PER LANGUAGE

Language	Refused: no pin → pinned	Harmful answer: no pin → pinned
PolyRefuse English	78% → 1%	**18% → 94%**
PolyRefuse German	68% → 11%	**26% → 89%**
PolyRefuse Spanish	66% → 5%	**28% → 86%**
PolyRefuse French	65% → 4%	**22% → 84%**
MultiJail English	59% → 7%	9% → 76%
MultiJail Chinese	57% → 8%	6% → 63%
MultiJail Italian	54% → 8%	12% → 68%
MultiJail Vietnamese	43% → 6%	16% → 60%

In every language so far, pinning one neuron cuts refusals to 1–11%. All 22 languages × 3 models done by tomorrow.

Inside the MLP: the refusal direction's top neuron is Apple's neuron

	Qwen3-8B	Llama-3.1-8B	Qwen3-1.7B
Direction's top neuron	**Apple's (23%)**	**Apple's (22%)**	**Apple's (18%)**
No intervention	13%	15%	38%
Remove the MLP direction	22%	23%	48%
Remove it, all neurons except Apple's	19%	17%	41%
Reset Apple's neuron to its harmless value	17%	23%	42%
Steer past harmless (α = 4)	**96%**	**92%**	**95%**
Pin Apple's neuron	**97%**	**94%**	**93%**
Nanda's ablation / with Apple's neuron held on	96% / 97%	96% / 97%	94% / 95%

% = harmful answers (attack success). Percent in row 1 = share of the direction's squared size on that one neuron.

- Reverse route works: build the direction first, and its biggest neuron is the refusal neuron, with no neuron search
- Switching refusal off is not enough; the neuron has to be pushed past its harmless value
- Nanda's direction does not act only through this neuron: holding it on does not bring refusal back

The MLP direction and the cross-language search agree

LLAMA-3.1-8B: THE MLP DIRECTION'S TOP NEURONS

Layer	Top neuron	Share	Recurs across languages?
10	F9064	11%	**yes, 13 of 22**
11	F4258 (Apple)	22%	**yes, 21 of 22**
15	F9635	10%	**yes, 18 of 22**

QWEN3-1.7B IN ARABIC

	PolyRefuse	MultiJail
Top neuron	**Apple's (23%)**	**Apple's (16%)**
No intervention	42%	16%
Remove direction	53%	20%
Steer, α = 4	81%	64%
Pin Apple's neuron	89%	58%

Two independent routes, one answer: searching neuron by neuron in 22 languages and decomposing Nanda's direction find the same small group of neurons. Arabic shows the same pattern.

Quick check: can the neuron also cause refusal?

Pinning the neuron to its harmless side removes refusal. Does pinning it to its harmful side make the model refuse harmless prompts?

100 HARMLESS PROMPTS, SHARE REFUSED

Model	Pin (toward harmful)	Refused: no pin → pinned
Qwen3-1.7B	−10 / −20 / −40	0% → 0% / 1% / 4%
Llama-3.1-8B	+1 / +2 / +4	**0% → 1% / 7% / 90%**
Qwen3-8B	−4 / −10 / −20	running

- Llama: pushing its one neuron to 4× its harmful value makes the model refuse 90% of harmless requests. The neuron both removes and creates refusal
- Qwen3-1.7B: refusals barely rise; there the neuron is needed for refusal but not enough on its own (it shares the job with L10:F244)
- Quick test: 100 prompts, short answers, string-match refusal check; to be confirmed with a judge

Next

This week

- Per-language attacks for all 3 models (running, done tomorrow)
- Apple's K.1 judge on those attacks
- Rerank in languages where the top neuron differs

Then

- Part 3 on Qwen3-8B and Llama in Arabic (queued)
- Yoruba / Swahili follow-up: restore refusal by pinning the neuron
- The neuron as one guard across languages

Later

- What the neuron responds to (Part 1 rows 9–10)
- Writing (December), ARR 4 January
- After: images and audio